Foundation holds given Regularity: minimal-rank witnesses break every nonempty set

The Open Logic Project contributors

openlogic:set-theory/spine:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-spine/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the optional argument '[working in $\ZFminus + \text{Regularity}$]' on the prop environment is part of the frozen source line; the preamble-side availability remark (lines 672-678) sits outside the span and is reflected in prereqs instead.

Editorial note. The label zfminusregularityfoundation names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Rank of an ordinal equals the ordinal; rank defined for all sets. Same unit, \ollabel{ordsetrankalpha} (source line 651) and the rank definition of the section. Lets 'B of least possible rank' be meaningful: ranks are ordinals and are well-ordered.
- Membership strictly lowers rank. Same unit, \ollabel{rankmemberslower} (source line 602). Line 685: c in B forces rank(c) in rank(B).

Source reference. Original reference rankmemberslower: sources/olp-set-theory-spine/pinned-original-source.tex, source lines 602–602. Prerequisite pointer invoked at proof line(s) 686; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1 (working in $\mathrm{ZF}^- + \text{Regularity}$)**.** *Foundation holds.*

Proof. Fix $A \neq \emptyset$, and some $B \in A$ of least possible rank. If $c \in B$ then $\operatorname{rank} c \in \operatorname{rank} B$ by rankmemberslower, so that $c \notin A$ by choice of B. □